Since A is a nonsingular matrix, A^{-1} exists

Now $\text{adj A} = \begin{bmatrix} 2 & -4 \\ 3 & 2 \end{bmatrix}$

$$A^{-1} = \frac{1}{|A|} adjA$$

$$= \frac{1}{16}\begin{bmatrix} 2 & -4 \\ 3 & 2 \end{bmatrix}$$

Example 1.15

If $\text{A} = \begin{bmatrix} -2 & 6 \\ 3 & -9 \end{bmatrix}$ then, find A^{-1}

Solution

$$|A| = \begin{vmatrix} -2 & 6 \\ 3 & -9 \end{vmatrix} = 0$$

Since A is a singular matrix, A^{-1} does not exist.

Example 1.16

If $\text{A} = \begin{bmatrix} 2 & 4 & 4 \\ 2 & 5 & 4 \\ 2 & 5 & 3 \end{bmatrix}$ then find A^{-1}

Solution

$$|A| = \begin{vmatrix} 2 & 4 & 4 \\ 2 & 5 & 4 \\ 2 & 5 & 3 \end{vmatrix}$$

$$= 2\begin{vmatrix} 5 & 4 \\ 5 & 3 \end{vmatrix} - 4\begin{vmatrix} 2 & 4 \\ 2 & 3 \end{vmatrix} + 4\begin{vmatrix} 2 & 5 \\ 2 & 5 \end{vmatrix}$$

$$= 2(15 - 20) - 4(6 - 8) + 4(10 - 10)$$

$$= 2(-5) - 4(-2) + 4(0)$$

$$= -10 + 8 + 0 = -2 \neq 0$$

Since A is a nonsingular matrix, A^{-1} exists.

$$A_{11} = -5;\ A_{21} = 8;\ A_{31} = -4$$

$$A_{12} = 2;\ A_{22} = -2;\ A_{32} = 0$$

$$A_{13} = 0;\ A_{23} = -2;\ A_{33} = 2$$

$$[A_{ij}] = \begin{bmatrix} -5 & 2 & 0 \\ 8 & -2 & -2 \\ -4 & 0 & 2 \end{bmatrix}$$

$$\text{adj A} = \begin{bmatrix} -5 & 8 & -4 \\ 2 & -2 & 0 \\ 0 & -2 & 2 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} adjA = \frac{1}{-2}\begin{bmatrix} -5 & 8 & -4 \\ 2 & -2 & 0 \\ 0 & -2 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{5}{2} & -4 & 2 \\ -1 & 1 & 0 \\ 0 & 1 & -1 \end{bmatrix}$$

Example 1.17

If $A = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$ satisfies the equation $A^2 - kA + I_2 = O$ then, find k and also A^{-1}.

Solution

$$\text{A} = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$$

$$A^2 = \text{A.A}$$

$$= \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 7 & 12 \\ 4 & 7 \end{bmatrix}$$

Given $A^2 - kA + I_2 = O$

$$\Rightarrow \begin{bmatrix} 7 & 12 \\ 4 & 7 \end{bmatrix} - k\begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 7 - 2k + 1 & 12 - 3k + 0 \\ 4 - k + 0 & 7 - 2k + 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 8 - 2k & 12 - 3k \\ 4 - k & 8 - 2k \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\Rightarrow 4 - k = 0$$

$$\Rightarrow \text{k} = 4.$$

Also $|A| = \begin{vmatrix} 2 & 3 \\ 1 & 2 \end{vmatrix}$

$$= 4 - 3 = 1 \neq 0$$

$$\text{adj } A = \begin{bmatrix} 2 & -3 \\ -1 & 2 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} adjA$$

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